

**LESSON 3-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**

# Understanding Ratios

Maryland Standard 6.RP.A.1 • Learning target – I can describe a relationship between two quantities using ratio language.

**REMEMBER**

- A **ratio** compares two quantities: 3 to 5, 3 : 5, or 3/5. **Order matters** – follow the order of the words.
- **Part-to-part** compares two groups. **Part-to-whole** compares one group to the total.
- **Simplest form**: divide both terms by their GCF.

**WORKED EXAMPLE**

A shelf has 10 science books and 15 history books. Write the ratio of science books to all the books in simplest form.

**Step 1** Total:  $10 + 15 = 25$  books

**Step 2** Science first: 10 : 25

**Step 3** Divide both terms by 5: 2 : 5

**Answer** **2 : 5 – for every 2 science books, there are 5 books in all**

**WORD BANK**

**ratio** – compares two amounts

**part-to-part** – one group to another group

**part-to-whole** – one group to the total

**simplest form** – no number divides both terms

**PRACTICE**

*show your work • label units*

1. A snack bag has 6 pretzels and 9 crackers. Write each ratio in simplest form.

**Step 1** Say the first group, then the second.

**Step 2** Divide both terms by the same number.

a. pretzels to crackers \_\_\_\_\_

b. crackers to pretzels \_\_\_\_\_

c. pretzels to all snacks \_\_\_\_\_

d. crackers to all snacks \_\_\_\_\_

2. Write each ratio in simplest form.

a.  $12 : 18 =$  \_\_\_\_\_

b.  $20 : 8 =$  \_\_\_\_\_

c.  $15 : 35 =$  \_\_\_\_\_

3. A team has **8 girls** and **7 boys**. Jada writes **8 : 15**. What does the **15** count?

**Frame** This ratio is part-to-\_\_\_\_\_ because 15 counts \_\_\_\_.

☐ Part-to-part

☐ Part-to-whole

**15 counts** \_\_\_\_\_

- 4. Class Trip Vote.** A class has **24 students**. **14** vote for the museum. The **rest** vote for the aquarium.

**Step 1** Find the aquarium votes:  $24 - 14$ .

**Step 2** Write museum : aquarium.

**Step 3** Divide both terms by the same number.

*Show your work.*

**Ratio** \_\_\_\_\_

- 5. Find the Error.** An animal shelter has 4 cats and 10 dogs. Marcus says the ratio of dogs to cats is 2 : 5. Explain his mistake and write the correct ratio.

**Frame** Marcus wrote the ratio in the wrong \_\_\_\_\_. Dogs to cats is \_\_\_\_ : \_\_\_\_, or \_\_\_\_ : \_\_\_\_ in simplest form.

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**CHECK YOURSELF**

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_

**AT HOME** Count two kinds of things at home, like forks and spoons. Say the ratio both ways.

**LESSON 3-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2****Ratios and Tape Diagrams**

Maryland Standard 6.RP.A.1 • Learning target – I can use a tape diagram to share a total according to a given ratio.

**REMEMBER**

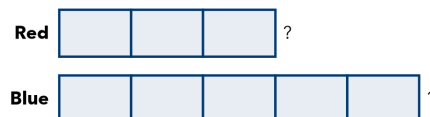
- A **tape diagram** uses equal boxes; the boxes match the ratio.
- **One box** =  $\text{total} \div \text{total parts}$ . Multiply one box by each term. Check that the amounts add to the total.

**WORKED EXAMPLE**

Leah and Omar share 35 stickers in the ratio 2 : 5. Find each share.

**Step 1** Total parts:  $2 + 5 = 7$ **Step 2** One box:  $35 \div 7 = 5$  stickers**Answer** Leah:  $2 \times 5 = 10$ . Omar:  $5 \times 5 = 25$ .**WORD BANK****tape diagram** – equal boxes for a ratio**total parts** – add the ratio terms**one box** –  $\text{total} \div \text{total parts}$ **PRACTICE***show your work • label units*

1. The tape diagram shows the ratio of red beads to blue beads on a bracelet. There are 48 beads in all.

**Step 1** Add the boxes:  $3 + 5$ .**Step 2** Divide 48 by the total boxes.**Step 3** Multiply one box by each term.**a.** What is the value of one box? \_\_\_\_\_ beads**b.** Find the red beads and the blue beads. \_\_\_\_\_

2. Find both amounts.

**a.** Share 42 in the ratio 3 : 4. \_\_\_\_\_ **b.** Share 66 in the ratio 2 : 9. \_\_\_\_\_

3. **Work Backward.** A fruit cup has grapes and melon pieces in the ratio 3 : 4. It has 12 grapes.

**Step 1** 12 grapes are 3 boxes. Find one box.**Step 2** Multiply one box by 4 for melon.**Step 3** Add grapes and melon.**a.** What is the value of one box? \_\_\_\_\_**b.** How many melon pieces? \_\_\_\_\_**c.** How many pieces of fruit are in the cup in all? \_\_\_\_\_

4. **Stretch.** Maya and Theo share stickers in the ratio **2 : 7**. Theo has **35 more** stickers than Maya.

**Step 1** Theo has  $7 - 2 = 5$  more boxes.

**Step 2** 5 boxes = 35. Find one box.

- a. What is the value of one box? \_\_\_\_\_ stickers  
b. How many stickers do they have in all? \_\_\_\_\_ stickers

5. **Find the Error.** Tyrell shared 45 marbles in the ratio 4 : 5 as 36 and 45. Explain his mistake. Find the correct shares.

**Frame** Tyrell divided by \_\_, but he should divide by the total parts, \_\_. Shares: \_\_ and \_\_.

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### TEST PRACTICE

*choose the best answer*

6. Jamal and Rosa share \$40 in the ratio 3 : 5. How much money does Rosa get?  
**A** \$5                      **B** \$15                      **C** \$24                      **D** \$25
7. A tape diagram shows a ratio of 4 : 3 with a total of 56. What is the value of one box?  
**A** 7                      **B** 8                      **C** 14                      **D** 32

#### CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_\_

**AT HOME** Show someone at home how to split 30 snacks in the ratio 2 : 3.